

Practice Problems Sheet VI

(Structure of Permutations)

Notation: We use S_n for the set of all permutations of the set $\{1, 2, \dots, n\}$.

Q1

- (a) Find the signs of permutations $\alpha = (1357)(2468)$ and $\beta = (135)(678)(2)(4)$.
- (b) How are signs of permutations τ and $\theta \circ \tau \circ \theta^{-1}$, where τ and θ are arbitrary elements of S_n , related?
- (c) Do permutations corresponding to same partition type have same sign?

Q2 Show that any cycle of odd length can be written as a product (not necessarily disjoint) of cycles of length 3. Also show that a permutation is even iff it can be written as a product of 3-cycles.

Q3 We say, $\sigma, \tau \in S_n$ are conjugates if $\exists \theta \in S_n [\sigma = \theta \circ \tau \circ \theta^{-1}]$.

- (a) Show that conjugacy is an equivalence relation on S_n .
- (b) Show that conjugate of a cycle of length r is a cycle of length r . More concretely, show that if $\sigma = \theta \circ \tau \circ \theta^{-1}$ and c_r is a cycle of length r in τ then $\theta(c_r)$ is a cycle of length r in σ .
- (c) Using part (b), show that if $\sigma, \tau \in S_n$ are conjugates then their cycle decompositions have the same partition type.
- (d) Show that if $\sigma, \tau \in S_n$ have the same partition type then they are conjugates.

All the parts above put together show that conjugacy relation on S_n is the same as having same partition type.

Q4

- (a) In class we derived a formula, using inclusion exclusion principle, for the number of surjective functions from an n -set to a k -set. Using this give an explicit formula, for the Stirling numbers of the second kind, $S(n, k)$.
- (b) Show that $S(n, k) = \frac{1}{k!} \sum \binom{n}{n_1, n_2, \dots, n_k}$, where sum is taken over all *positive* integers satisfying $n_1 + n_2 + \dots + n_k = n$.

Q5 For a permutation $\alpha \in S_6$ of type $[1^4 2]$, how many $\beta \in S_6$ are there s.t. $\alpha \circ \beta = \beta \circ \alpha$?
How many for α of type $[2^3]$?

Q6 List the *types* of permutation in S_7 which are derangements, and hence check the value of d_7 against the value obtained from inclusion exclusion formula.

End